ALGEBRA 2 · LINEAR FUNCTIONS & SYSTEMS

Feasible Region Optimizer

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Feasible Region Optimizer** on Mr. K's Classroom: Algebra 2 → Linear Functions & Systems → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. Where in the shaded region do you think the best profit happens: the middle, an edge or a corner?

   At a corner (vertex).

2 Explore use the tool and record what you see

Profit P = 30x + 20y (oven hours = 16)

Corner (x, y)	(0, 0)	(8, 0)	(6, 4)	(2, 8)	(0, 8)
Profit	**$0**	**$240**	**$260**	**$220**	**$160**

Change the profits

Profit function	Best corner	Best profit
P = 30x + 20y	**(6, 4)**	**$260**
P = 10x + 20y	**(2, 8)**	**$180**
P = 40x + 20y	**(8, 0) and (6, 4) tie**	**$320**

3 Write the rule fill in each blank from what you observed

1. The feasible region contains every point that satisfies **all of** the constraints.
2. The maximum or minimum of the objective function happens at a **vertex (corner)** of the region.
3. When the profit line is parallel to an edge of the region, **two** corners can tie.
4. Sliding the profit line farther from the origin **increases** the profit.

4 Check no tool: show your work

1. A region has corners (0, 0), (0, 6), (4, 4) and (7, 0). Maximize C = 5x + 3y.

   35 at (7, 0)

2. Using the same corners, minimize C = 5x + 3y.

   0 at (0, 0)

3. Is (3, 5) in the bakery region (x + y ≤ 10, 2x + y ≤ 16, y ≤ 8)?

   Yes: 8 ≤ 10, 11 ≤ 16, 5 ≤ 8

4. Why don't we need to test points inside the region?

   A linear objective is always larger at some corner than at any inside point

5 Reflect

Explain why the best value must be at a corner.

The profit line keeps moving in one direction as P grows; the last point of the region it touches is a corner.